

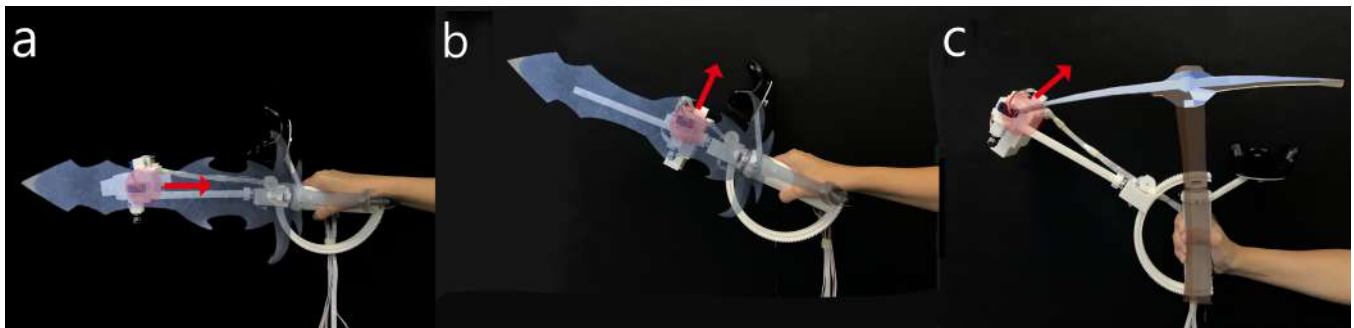
# Demonstration of transPAF: Rendering Omnidirectional Impact Feedback with Dynamic Point of Application of Force All Round a Controller

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**Figure 1:** transPAF dynamically changes the points of application and direction of force to render realistic impact force for versatile scenarios. (a), (b), and (c) show the PAF and the force direction of stabbing the sword, slashing the sword, and the pickaxe, respectively.

## ABSTRACT

Impact force is common haptic feedback on virtual reality (VR) controllers, such as hitting objects with weapons or rackets. It applies to different points of application of force (PAFs) and directions in varied scenarios. Therefore, we propose a controller, transPAF, to render omnidirectional impact feedback with dynamic PAF all round the controller for versatile VR scenarios. transPAF consists of a controller, a semicircular track, a linear track, and an impactor, which are all rotatable. The impactor can move to any position in a sphere, which means the whole 3D space all round the controller, and rotate in any direction. In the demonstration, users can use four weapons, including a sword, a pickaxe, a hook, a dagger, and a tennis racket in the VR scene to attack monsters and hit balls, so that they can feel the sensation of the impact force applied to different PAFs and directions.

## CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; **Haptic devices**.

## KEYWORDS

Haptic feedback, impact force feedback, point of application of force, virtual reality.

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## 1 INTRODUCTION

Haptic feedback plays an important role in virtual reality (VR) realism. For various haptic feedback, impact force is a common one applying to a VR controller, such as hitting objects with a weapon or racket. Impact force applies to different points of application of force (PAFs) [5] and in varied directions in different scenarios. For example, due to different shapes of a sword and a pickaxe, where these weapons hit objects are different, which causes varied PAFs. Furthermore, using a weapon in different ways also causes

various PAFs and force directions, such as stabbing or slashing with a sword. In addition, when playing tennis, a ball flies from different directions to hit the racket, causing different impact force directions. Therefore, to achieve realistic VR experiences, rendering dynamic PAF and force direction is essential.

To render dynamic PAF, some methods [2, 3, 6] utilize weight-shifting mechanisms to simulate weight change. However, these only use the property of dynamic PAF of gravity, and the force direction is constantly toward the ground, which are limited. Aero-plane [1] uses propellers to simulate dynamic PAF and generate the sensation of an object moving on a 2D plane. ElastiLinks [5] first investigates the essential of dynamic PAF in 3D space. However, it focuses on force feedback between controllers, the PAF is only on the surface of a cylinder around the controller, and the force direction depends on the other controller, which limit the position and direction of the dynamic PAF. Therefore, force feedback with dynamic PAF and dynamic force direction is still not accomplished.

We propose a controller, transPAF, to render omnidirectional impact feedback with dynamic PAF all round the controller for versatile VR scenarios. transPAF is composed of a controller, a semi-circular track, a linear track and an impactor, which are all rotatable. The controller rotates the semi-circular track, the linear track moves along the semi-circular track, and the impactor moves along the linear track. Therefore, the impactor can move to any position in a sphere around the controller. Such a high degree-of-freedom (DoF) design covers almost the whole 3D space all round the controller. Furthermore, by spinning the linear track and rotating the impactor, the impactor can render omnidirectional impact force. We further demonstrate five VR applications, including a sword, a pickaxe, a hook, a dagger, and a tennis racket in the VR scene to attack monsters and hit balls, to provide users the sensation of impact force applied to different PAFs and directions.

## 2 TRANSPAF

To render impact feedback on a controller with dynamic PAF and force direction to simulate handheld objects with different shapes hitting other objects in different directions, we propose transPAF. transPAF consists of a controller, a semi-circular track, a linear track, and an impactor. All components are rotatable. Therefore, by leveraging the combinations of moving and rotating the tracks and the impactor, transPAF achieves dynamic PAF and force direction. The controller is an elliptical cylinder. A DC motor and a bearing are embedded in the upper and lower ends of the controller, respectively, to rotate the semi-circular track. Since we intend to move the impactor to any position all round a sphere for dynamic PAF, the controller rotates the semi-circular track on the theta plane for the azimuthal angle in the spherical coordinate system. The controller can render  $\theta$  angle range from  $-150^\circ$  to  $150^\circ$ .

The linear track moves along the semi-circular track by leveraging a DC motor. The linear track moves on the phi plane for the inclination or polar angle in the spherical coordinate system, and the  $\phi$  angle range is from  $-90^\circ$  to  $90^\circ$ . The impactor moves along the linear track using a DC motor in a range of 15cm. The rotation and movement of the controller, semi-circular track, linear track and impactor are corresponding to the azimuthal angle ( $\theta$ ), inclination angle ( $\phi$ ) and radial distance in the spherical coordinate system.

Therefore, transPAF can move the impactor to any position in a sphere around the controller.

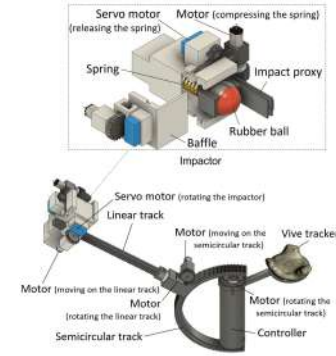


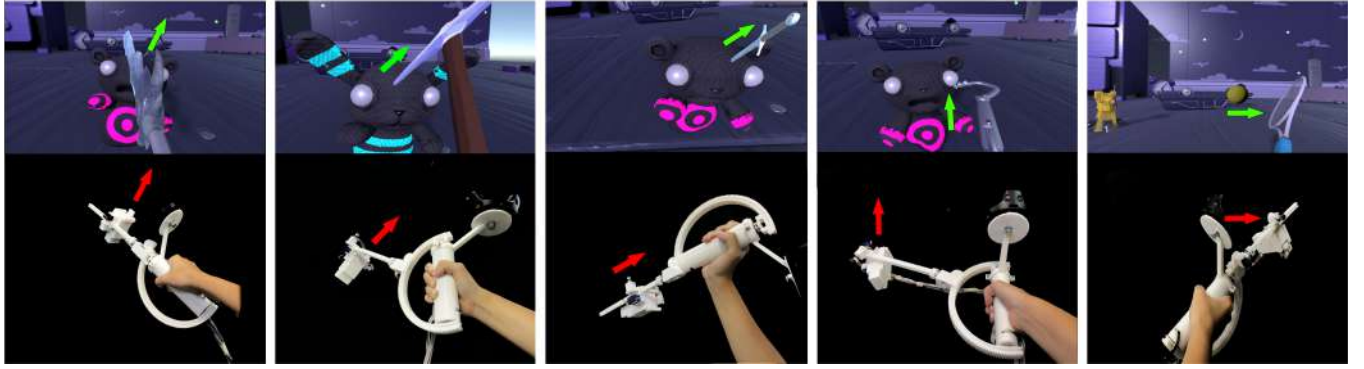
Figure 2: The hardware structure of the transPAF prototype.

To render omnidirectional impact, the impactor should rotate around two axes. The linear track not only moves along but also spins on the semi-circular track using another DC motor embedded in it in a range of  $-180^\circ$  to  $180^\circ$  on the theta plane in the azimuthal ( $\theta_r$ ) angle. The impactor rotates in a range of  $-90^\circ$  to  $90^\circ$  in the inclination ( $\phi_r$ ) angle by a servo motor. This mechanism allows the impactor to render omnidirectional impact force. Since the impactor position control, including the  $\theta$ ,  $\phi$ , offset, and direction control, including  $\theta_r$  and  $\phi_r$ , are independent, dynamic PAF and force direction are accomplished.

For the impactor, we based on and improved the design in [4] to leverage elastic or spring force to generate instant impact. It consists of a compression spring, motors, rack and pinion mechanisms, a barrel and an impact proxy. The compression spring and impact proxy are in the barrel. One end of the spring connects to the end of the barrel, and the other connects to the impact proxy. One rack and pinion mechanism is used to move the impact proxy to compress the spring. The rack is attached to the impact proxy, which is further driven by the pinion using a DC motor. The other rack and pinion mechanism, driven by a servo motor, is used to move the DC motor driving the impact proxy. Therefore, when the DC motor and its pinion are lifted, the pinion disengages the rack of the impact proxy and releases the compressed spring. Impact is then generated when the impact proxy hits the baffle on the other end of the barrel. After that, the DC motor and its pinion are moved back to re-engage and drive to the rack of the impact proxy and compress the spring again to store impact power. By compressing the spring at different distances, the multilevel impact is rendered.

## 3 APPLICATION

Users use four weapons, including a sword, a pickaxe, a hook and a dagger, and a tennis racket in the VR scene to attack monsters and hit balls, respectively. For the sword, there were two ways to use it, slashing and stabbing. For slashing with the sword, according to the position of the blade of the sword hitting monsters, the PAF could be in three positions. Therefore, from the tip to the handle of the sword, the PAF and force direction ( $\theta$ ,  $\phi$ , offset,  $\theta_r$ ,  $\phi_r$ ) were  $(0^\circ, 90^\circ, 150, 0^\circ, 0^\circ)$ ,  $(0^\circ, 90^\circ, 50, 0^\circ, 0^\circ)$  and  $(0^\circ, 90^\circ, -50, 0^\circ, 0^\circ)$ .



**Figure 3: Users use four weapons, including a sword, a pickaxe, a hook and a dagger, and a tennis racket in the VR scene to attack monsters and hit balls, respectively**

For stabbing with the sword, the tip of the sword hit monsters, so the PAF and force direction  $(\theta, \phi, \text{offset}, \theta_r, \phi_r)$  were  $(0^\circ, 90^\circ, 150, 0^\circ, 90^\circ)$ .

For the pickaxe, since it was in the ‘T’ shape, the PAF and force direction  $(\theta, \phi, \text{offset}, \theta_r, \phi_r)$  were  $(0^\circ, 30^\circ, 150, 0^\circ, 0^\circ)$ .

For the hook, since it was curved, the tip was inward and on the left side of the handle when it was held with a right hand. Its PAF and force direction  $(\theta, \phi, \text{offset}, \theta_r, \phi_r)$  were  $(-90^\circ, 45^\circ, 150, 0^\circ, -45^\circ)$ . For the dagger, it was like a small sword or knife held with the reverse grip, and it could only be used to stab monsters. Therefore, its PAF and force direction  $(\theta, \phi, \text{offset}, \theta_r, \phi_r)$  were  $(0^\circ, -90^\circ, 50, 0^\circ, 90^\circ)$ .

For the racket, when a ball hit the racket at the sweet spot in the forehand and backhand, the PAF and force direction  $(\theta, \phi, \text{offset}, \theta_r, \phi_r)$  were  $(0^\circ, 90^\circ, 70, 90^\circ, 0^\circ)$  and  $(0^\circ, 90^\circ, 70, -90^\circ, 0^\circ)$ , respectively. Three turrets in the VR scene shot tennis balls to the racket, so the racket might hit balls in different directions. Therefore, the impactor direction was adjusted correspondingly. In most of these scenarios, the impact force magnitude was 4.5N. However, when using the dagger and slower balls were shot, the impact force was 3N.

In the application, when switching to a weapon or racket, the PAF and force direction could be changed correspondingly before using the weapon or racket. Furthermore, the impactor could also store impact power in advance. When the weapons or racket contacted monsters or balls, the impactor only needed to render impact.

## 4 CONCLUSION

We propose a controller, transPAF, to render omnidirectional impact feedback with dynamic PAF all round the controller. transPAF is composed of a controller, a semicircular track, a linear track and an impactor, which are all rotatable. By moving these tracks, the impactor can move to any position in a sphere around the controller. Furthermore, by spinning the linear track and rotating the impactor, the impactor can render omnidirectional impact force. We further demonstrate five VR applications, including a sword, a pickaxe, a hook, a dagger, and a tennis racket in the VR scene to attack monsters and hit balls, to provide users the sensation of impact force applied to different PAFs and directions.

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